Department of Mechanical Engineering
University of South Carolina
Columbia, SC 29208
January 30, 2026

Dear Members of the Evaluation Committee:

I recommend Mateo Altomare for the SC Space Grant Undergraduate Research Award. As his faculty advisor in the Integrated Multiphysics and Systems Engineering Laboratory (iMSEL) since June 2025, I have supervised his research on multi-spectral sensor fusion for autonomous maritime perception.

Mateo joined iMSEL as a rising sophomore with no prior experience in machine learning or computer vision. He progressed rapidly from introductory Python to implementing camera calibration routines for our thermal-visible sensor rig, and was independently writing PyTorch training scripts for our Detection Transformer (DETR) model on maritime imagery within his first semester.

He is now co-author on a paper planned for submission at ASME IDETC 2026 presenting a method for pixel-level infrared-to-visible image registration, having derived and implemented the full registration pipeline end-to-end.

Three accomplishments illustrate his technical range. First, he deployed real-time object detection on a power-constrained Raspberry Pi for automated target identification in marine radar displays. Second, he built our multi-spectral detection pipeline that fuses thermal and visible imagery for cross-modal calibration, which now generates all training data for his proposed framework. Third, he curated and annotated approximately 750 thermal-visible frame pairs spanning 1–75 m depth range with a structured labeling protocol tracking scene conditions, object depth, and registration quality.

His proposed research – Learned Depth-Adaptive Registration (LDAR) – addresses a real limitation in our current workflow, which requires knowing object depth in advance. Mateo's proposal would automate depth estimation, eliminating manual specification and additional hardware. I have reviewed his proposed architecture in detail and find the approach to be technically rigorous, well-matched to the available dataset, and targeting an accuracy threshold that, while ambitious, is well within reach based on our preliminary results.

Beyond research, Mateo co-leads the Gamecock Motorsports Formula SAE chassis team and maintains a 3.91 GPA. Based on his completion of all Fall 2025 milestones ahead of schedule, I am confident he will execute the proposed plan and deliver the stated outcomes.

I give Mateo my highest recommendation for this award.

Sincerely,

Dr. Yi Wang
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